

SINAIA, Romania

WMO: 153250

Lat: 45.3550N

Lon: 25.5142E

Elev: 1511

StdP: 84.44

Time Zone: 2.00 (EUE)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-16.2	-14.1	-21.1	0.7	-10.5	-18.0	0.9	-9.6	18.1	-9.3	14.1	-8.4	2.0	0	0.611

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	7.7	23.2	16.4	21.7	15.3	20.3	14.7	17.3	22.1	16.4	20.6	15.5	19.4	1.0	20

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
15.6	13.3	19.7	14.7	12.6	18.7	14.0	12.0	17.9	55.0	22.4	51.9	20.6	49.1	19.5	20.4

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
11.5	8.3	6.5	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	5.3	-4.9	-2.3	-0.9	3.7	9.1	13.5	14.2	14.8	10.3	5.7	1.9	-2.5
	DBStd	8.26	5.50	4.91	4.86	5.69	3.84	3.59	3.40	3.65	4.92	5.24	5.13	4.98
	HDD10.0	2271	462	346	338	198	62	11	6	4	59	155	244	386
	HDD18.3	4789	720	579	597	438	286	149	131	117	244	391	493	644
	CDD10.0	541	0	0	0	10	34	115	138	153	66	22	2	0
	CDD18.3	17	0	0	0	0	3	4	7	2	0	0	0	0
	CDH23.3	48	0	0	0	0	0	11	13	18	6	0	0	0
	CDH26.7	2	0	0	0	0	0	1	1	1	0	0	0	0
Wind	WSAvg	2.2	2.5	2.4	2.6	2.4	2.0	2.0	2.0	2.0	2.0	2.1	2.2	2.2
Precipitation	PrecAvg	883	44	40	51	71	104	123	114	92	74	66	52	53
	PrecMax	1242	93	88	96	159	207	204	227	237	142	132	111	141
	PrecMin	537	9	6	13	34	26	47	38	26	25	5	3	7
	PrecStd	174	23	19	22	32	47	47	48	50	37	32	27	31
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	9.7	10.0	13.0	18.1	20.4	24.8	24.8	25.3	23.4	19.4	15.2	12.2
		MCWB	2.7	4.9	6.2	10.4	12.5	17.8	16.9	17.2	15.9	11.3	7.7	4.0
	2%	DB	5.5	7.9	10.1	16.0	18.6	22.2	22.9	23.1	20.6	17.4	12.1	7.9
		MCWB	2.0	3.5	4.6	9.2	12.4	15.8	16.7	16.1	13.9	10.3	6.6	2.2
	5%	DB	3.8	5.9	7.9	13.7	16.7	20.4	21.0	21.6	18.5	14.9	10.0	5.5
		MCWB	0.9	2.2	3.4	8.0	11.6	15.2	15.6	15.3	12.8	9.1	5.8	1.8
	10%	DB	2.1	4.2	5.9	11.4	14.8	18.8	19.3	19.9	16.5	12.7	8.3	3.6
		MCWB	0.0	1.1	2.4	6.7	10.6	14.2	14.7	14.7	12.0	7.9	4.6	1.0
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	3.7	6.0	6.7	11.5	14.0	18.4	18.2	18.5	16.2	12.9	8.8	5.0
		MCDB	6.6	9.0	12.1	16.3	18.8	22.6	23.4	24.1	22.5	17.7	12.5	8.5
	2%	WB	2.7	4.1	5.3	9.9	12.8	16.7	17.1	17.2	14.4	11.4	7.8	3.7
		MCDB	4.9	7.0	9.4	15.3	17.2	21.3	21.7	22.0	19.8	15.6	10.6	6.3
	5%	WB	1.6	2.8	3.9	8.7	12.0	15.7	16.2	16.2	13.5	10.1	6.9	2.6
		MCDB	3.6	5.2	7.2	13.0	15.9	19.6	20.2	20.4	18.0	14.1	9.4	4.9
	10%	WB	0.1	1.7	2.7	7.2	10.9	14.8	15.3	15.3	12.4	8.8	5.8	1.5
		MCDB	1.7	3.9	5.5	10.9	14.4	18.4	18.7	19.4	16.2	12.0	8.1	3.5
Mean Daily Temperature Range	5% DB	MDBR	6.4	6.8	6.9	7.5	7.3	7.8	7.7	7.7	6.9	6.7	6.0	6.0
		MCDBR	8.4	8.4	9.5	10.1	9.5	9.8	9.4	9.2	9.2	9.4	8.0	7.8
		MCWBR	5.6	5.6	5.8	6.2	6.3	6.3	6.2	5.8	5.9	6.7	5.6	5.1
	5% WB	MCDBR	7.5	7.6	8.7	9.4	9.1	9.4	8.9	8.9	8.7	9.2	6.2	6.8
		MCWBR	5.3	5.4	5.8	5.8	6.2	6.3	6.2	5.9	5.7	6.8	5.0	4.9
Clear-Sky Solar Irradiance	taub	0.264	0.288	0.322	0.370	0.355	0.357	0.368	0.380	0.344	0.315	0.293	0.267	
	taud	2.400	2.349	2.344	2.261	2.353	2.366	2.366	2.334	2.298	2.387	2.472	2.489	2.444
	Ebn at Noon	860	897	909	887	912	908	893	868	874	858	818	819	
	Edh at Noon	71	92	107	127	120	119	122	121	102	81	66	62	
All-Sky Solar Radiation	RadAvg	1.40	2.22	3.31	4.33	5.29	5.72	5.82	5.25	3.85	2.64	1.54	1.14	
	RadStd	0.15	0.29	0.35	0.47	0.51	0.51	0.48	0.42	0.36	0.34	0.16	0.14	

Historical Trends

	DBAvg	Heating			Cooling			Degree-Days			
		99% DB		99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (73 neighbors)	+0.55	N/A	N/A	+0.66	N/A	N/A	-115	-161	+72	+24

Nomenclature: See separate page